# $\mathrm{SP}_N$ and $\mathrm{SDP}_N$ Verification and Validation

Brantley–Larsen and C5G7 against Carreño et al. (2024)

`GARNET` V&V report, audit item O10

30 October 2026

## Summary

This report verifies `GARNET`'s $\mathrm{SP}_N$ and $\mathrm{SDP}_N$ solvers against the two two-dimensional problems of Carreño et al. (2024), the paper whose equations they implement, after the October 2026 code audit (item O10).

- **Brantley–Larsen, code verification.** With the paper's coupled Marshak vacuum, `GARNET`'s mesh limits equal the paper's Table 6 within 0.6 pcm for SP1, SP3, SP5, SP7, SDP1 and SDP2. Convergence is second order (observed 1.93–2.00), and the extrapolation is good to about 0.2 pcm.
- **SDP3 coefficient tables.** The paper's SDP3 results were computed with the tables in the authors' FEMFFUSION code: with them, `GARNET` reproduces Table 6 within 0.2 pcm. Those tables differ from the paper's printed Eq. (A.6) (which gives +24 pcm) and from the first-principles tables `GARNET` uses (−10 pcm on Brantley–Larsen, about −14 pcm on C5G7).
- **C5G7, pin-resolved.** The pin powers reproduce the paper's Table 7 AVG and RMSE against the NEA MCNP reference within 0.04 percentage points (AVG 1.43% for diffusion down to 0.53–0.57% for SP5, SP7 and SDP2). The eigenvalues are 11–22 pcm below the paper's (42 pcm for SDP3 with `GARNET`'s tables), a model difference of 3–5% of the angular error whose source is not identified. Both codes show the paper's C5G7 conclusion: $k$ stagnates with angular order while the pin powers improve.
- **Flux shapes.** Along the lines the paper plots, `GARNET` agrees with the OpenMOC and OpenMC profiles as closely as the paper's own solutions do (0.4% of the peak on Brantley–Larsen, 1.1–1.7% on C5G7 for SDP3).
- **Findings for the code** (Section 7): the per-moment Robin vacuum, then the default of every solver except SP3, is 52–73 pcm low on a leaky core, so Marshak is now the default wherever it is available, with Robin kept for comparison; the C5G7 builder's sampled pin areas misstated the fuel inventory and moved $k$ by up to 226 pcm, so exact areas are now the default; and the Jacobi-preconditioned inner solves hit their iteration cap on the finest 2-D meshes (GMRES from 1.6 million cells, CG at 6.6 million).
- **GPU against CPU.** On identical meshes, one GH200 is faster than one CPU core from about $10^5$ unknowns and 22–162 times faster on the largest 2-D meshes both ran. It solves 3-D-sized, extruded versions of both problems with up to 597 million unknowns in at most 34 minutes; on those with 64 layers it is 350–400 times faster than one core for SP1 and SP3 (measured), and about 200–260 times for SDP1 and SDP3 (projected). *Interim:* two single-core runs and two controls were still running on 2 November; Section 5 will be updated when they finish.
- **Tests.** The Brantley–Larsen mesh-limit test now covers every method at 4 pcm (was SP3 and SDP1 at 10 pcm); a slow pin-resolved C5G7 test checks diffusion against the paper's $k$ and pin powers.

# 1 Scope and references

The reference is A. Carreño, A. Vidal-Ferràndiz, D. Ginestar and G. Verdú, "Development of a diffusive simplified double $P_N$ approximation to the neutron transport equation", *Ann. Nucl. Energy* 207 (2024) 110675 (CC BY 4.0, in `literature/`). `GARNET`'s $SP_N$ ($N = 1, 3, 5, 7$) and $SDP_N$ ($N = 1, 2, 3$) solvers implement the same equations in the same unified form. The paper solves them with second-degree finite elements; `GARNET` uses cell-centred finite volumes. Two of the paper's problems are used:

- **Brantley–Larsen** (paper Sec. 4.2): one group, three fuel bars in a $10 \times 10$ cm moderator square, reflective at $x = 0$ and $y = 0$, vacuum at $x = 10$ and $y = 10$ (paper Fig. 3, Table 5). References: the paper's own $SP_N$/$SDP_N$ eigenvalues (Table 6), OpenMOC ($k = 0.80536$) and TWODANT $S_{16}$ ($k = 0.80613$), and the OpenMOC flux along $y = 4.5$ cm (Fig. 4).
- **C5G7** (paper Sec. 4.3): the OECD/NEA two-dimensional quarter core, seven groups, vacuum outer boundary (Fig. 5). References: the paper's eigenvalues and pin-power errors (Table 7), MCNP ($k = 1.18655$), the NEA MCNP pin powers, and the OpenMC flux along two assembly diagonals (Figs. 6–7).

Three questions are separated. *Solution verification*: how far each `GARNET` eigenvalue is from its own mesh limit. *Code verification*: whether the mesh-converged `GARNET` result equals the paper's solution of the same equations. *Validation of the approximation*: how far $SP_N$ and $SDP_N$ are from transport. The last is a property of the equations, so `GARNET` should reproduce the paper's distance from transport rather than improve on it.

# 2 Code, models and metrics

**Code state.** Commit `20967d3` (the merged 2026-10-28 audit) plus the uncommitted audit work of 29–30 October. The audit changes that bear on these problems are: the coupled Marshak vacuum is now the default of the dedicated SP3 solver; plain power iteration stops on an error bound rather than the last change; non-symmetric $SDP_N$ blocks default to GMRES; and, added for this study, `build_c5g7_2d` accepts exact pin areas (`exact_area`) and a vacuum outer boundary (`outer_bc="vacuum"`). Following this study (finding 4), exact areas are now the default of the pin-resolved C5G7 and C5G7-TD builders; every run here names its area rule explicitly, so that change moves none of the results.

**Discretization.** Cell-centred finite volumes on a uniform Cartesian mesh, second order where material interfaces lie on cell faces. In the pin-resolved C5G7 model a cell cut by the $r = 0.54$ cm fuel circle is a volume blend of pin and water: linear in the reaction cross sections, harmonic in the diffusion coefficients, and weighted by fission production in the emission spectrum. Its covered fraction is computed in closed form. The pin-cell-homogenized model, which `tests/validation/test_c5g7.py` runs, volume-weights each 1.26 cm pin cell instead.

**Vacuum boundaries.** *Marshak* is the coupled condition $-\mathbf{n} \cdot D\nabla U = (g \otimes I)\, U$ that the paper uses. *Robin* applies $\alpha = 1/2$ to each moment separately. SP1 is diffusion, for which the two coincide. When these runs were made only the dedicated SP3 solver defaulted to Marshak; following this study (finding 3), every $SP_N$/$SDP_N$ eigen, transient and noise solver now does so wherever the coupled term exists, and Robin stays available as `marshak_vacuum=False`. The dedicated SDP1 solver has no Marshak term and keeps Robin; its Marshak runs use the general $SDP_N$ block of order 1. Every run here names its boundary explicitly, so the change of default moves none of the results.

**Tolerances.** Brantley–Larsen: `tol_k` $= 10^{-10}$, `tol_source` $= 10^{-9}$. C5G7: $10^{-9}$ and $10^{-8}$. Every run converged.

**Metrics.** Eigenvalue differences are signed, $10^5(k - k_{\mathrm{ref}})$ in pcm. The paper's Eq. (49) defines a relative difference, but its Tables 6 and 7 list $10^5|k - k_{\mathrm{ref}}|$ (for example SP1 in Table 7: $10^5 \times |1.18336 - 1.18655| = 319$); the tables are what is compared here. Pin-power errors $e_i = (P_i - P_i^{\mathrm{ref}})/P_i^{\mathrm{ref}}$ over the 1056 fuel pins, with powers normalized to a mean of 1: AVG $= \langle |e_i| \rangle$ (paper Eq. 50); the paper's RMSE, whose Table 7 values match $\sqrt{\sum e_i^2}/N$ (Eq. 51 as printed equals AVG); the conventional RMS $\sqrt{\langle e_i^2 \rangle}$; MRE $= \sum |e_i| P_i^{\mathrm{ref}} / \sum P_i^{\mathrm{ref}}$; and the largest $|e_i|$. Flux profiles are compared after a least-squares scale factor (the paper's flux units are arbitrary), as RMS and largest deviation in percent of the reference peak.

**Paper figures.** The paper's plots are vector graphics, so the reference curves were read from the drawn paths themselves (`extract_paper_figures.py`), not digitized from pixels. Axis calibration is a linear fit through the tick marks; the residual plus half the line width is below 0.4% of the curve's peak in all five panels.

# 3 Brantley–Larsen

## 3.1 Problem and runs

One energy group: fuel $\Sigma_t = 1.5$, $\Sigma_{s0} = 1.35$, $\nu\Sigma_f = 0.24$ cm$^{-1}$; moderator $\Sigma_t = 1.0$, $\Sigma_{s0} = 0.93$ cm$^{-1}$ (paper Table 5). The fuel bars occupy $x \in [1,2] \cup [4,5] \cup [7,8]$, $y \in [0,9]$ cm. Every interface lies on a cell face for $n$ a multiple of 10, so the finite-volume scheme is second order. Each method was run with Marshak and Robin vacuum on $n \times n$ meshes, $n = 40, 80, 160, 320, 640$. The GMRES runs of SDP2 (Marshak) and SDP3 (both boundaries) at $n = 640$ would have taken 8–20 h each on the shared node and were stopped; for those three the extrapolation uses $n = 80, 160, 320$.

## 3.2 Mesh convergence

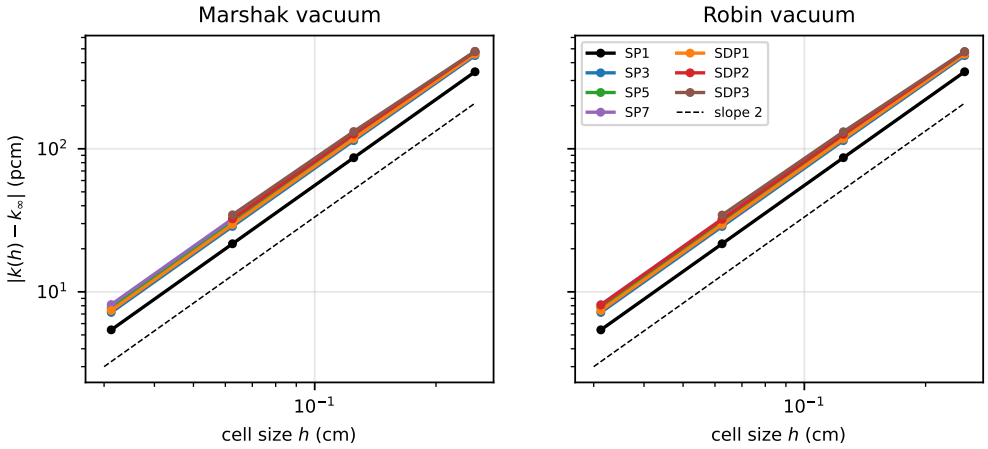



Figure 1: Brantley–Larsen: distance of each mesh's $k$ from its own extrapolated limit. Every method and boundary treatment converges at second order.

Table 1 and Fig. 1. The observed order from the finest three meshes is between 1.93 and 2.00 for every method and both vacuum treatments, and $k$ decreases monotonically under refinement. $k_\infty$ is the Richardson extrapolation with the observed order. The GCI column is Roache's grid-convergence index with safety factor 1.25; it is 1.7–2.6 pcm where $n = 640$ exists and 10–11 pcm where the triple ends at $n = 320$. The GCI is a conservative bound: wherever both triples exist (14 configurations), extrapolating from $n = 80, 160, 320$ and from $n = 160, 320, 640$ agrees within 0.21 pcm.

On the $80 \times 80$ mesh of `tests/validation/test_brantley_larsen_2d.py` the mesh error is +87 pcm for SP1 and +114 to +132 pcm for the transport methods (Table 3). That is larger than the difference between consecutive SP$_N$ orders above SP5, so single-mesh comparisons with the paper on that mesh mostly measure mesh error.

## 3.3 Comparison with the paper

With the Marshak boundary, `GARNET`'s mesh limit equals the paper's Table 6 within 0.6 pcm for SP1, SP3, SP5, SP7, SDP1 and SDP2 (Table 2, Fig. 2), about the ±0.5 pcm rounding of Table 6. This is the code verification: the same equations, discretized independently (finite volumes here, second-degree finite elements in the paper), give the same eigenvalue.

Table 1: Brantley–Larsen eigenvalues by mesh, observed order $p$, extrapolated $k_\infty$ and GCI.

| Method | Vacuum | $k_{80}$ | $k_{160}$ | $k_{320}$ | $k_{640}$ | $p$ | $k_\infty$ | GCI (pcm) |
|---|---|---|---|---|---|---|---|---|
| SP1 | – | 0.777667 | 0.777018 | 0.776855 | 0.776814 | 2.00 | 0.776801 | 1.70 |
| SP3 | Marshak | 0.800178 | 0.799323 | 0.799109 | 0.799055 | 2.00 | 0.799037 | 2.25 |
| | Robin | 0.799648 | 0.798795 | 0.798580 | 0.798526 | 2.00 | 0.798508 | 2.25 |
| SP5 | Marshak | 0.804023 | 0.803112 | 0.802881 | 0.802823 | 1.99 | 0.802803 | 2.43 |
| | Robin | 0.803488 | 0.802578 | 0.802347 | 0.802289 | 1.99 | 0.802270 | 2.43 |
| SP7 | Marshak | 0.804814 | 0.803869 | 0.803627 | 0.803566 | 1.99 | 0.803546 | 2.57 |
| | Robin | 0.804273 | 0.803331 | 0.803089 | 0.803028 | 1.99 | 0.803007 | 2.56 |
| SDP1 | Marshak | 0.802787 | 0.801908 | 0.801687 | 0.801631 | 2.00 | 0.801613 | 2.32 |
| | Robin | 0.802056 | 0.801178 | 0.800957 | 0.800901 | 2.00 | 0.800883 | 2.32 |
| SDP2 | Marshak | 0.804992 | 0.804050 | 0.803809 | – | 1.97 | 0.803726 | 10.36 |
| | Robin | 0.804464 | 0.803524 | 0.803284 | 0.803223 | 1.99 | 0.803203 | 2.54 |
| SDP3 | Marshak | 0.805241 | 0.804266 | 0.804011 | – | 1.93 | 0.803921 | 11.30 |
| | Robin | 0.804699 | 0.803728 | 0.803474 | – | 1.93 | 0.803383 | 11.26 |

Table 2: Brantley–Larsen: extrapolated `GARNET` eigenvalues against the paper's Table 6 and OpenMOC (pcm). "Robin − paper" uses `GARNET`'s Robin-vacuum $k_\infty$.

| Method | Paper $k$ | Paper − OpenMOC | `GARNET` $k_\infty$ | − paper | Robin − paper | `GARNET` − OpenMOC |
|---|---|---|---|---|---|---|
| SP1 | 0.77680 | -2856 | 0.77680 | +0.1 | – | -2856 |
| SP3 | 0.79904 | -632 | 0.79904 | -0.3 | -53.2 | -632 |
| SP5 | 0.80280 | -256 | 0.80280 | +0.3 | -53.0 | -256 |
| SP7 | 0.80354 | -182 | 0.80355 | +0.6 | -53.3 | -181 |
| SDP1 | 0.80161 | -375 | 0.80161 | +0.3 | -72.7 | -375 |
| SDP2 | 0.80373 | -163 | 0.80373 | -0.4 | -52.7 | -163 |
| SDP3 | 0.80402 | -134 | 0.80392 | -9.9 | -63.7 | -144 |

SDP3 is −9.9 pcm from the paper. `GARNET` does not use the SDP3 coefficient rows 2–3 printed in the paper's Eq. (A.6): a from-scratch symbolic derivation from the paper's own closure reproduces its SDP1 and SDP2 tables and every Marshak matrix exactly, but not those rows (`tests/verification/test_sdpn_derivation.py`, note in `garnet/spn.py`). Section 3.7 shows that the paper's SDP3 values come from a third set, the tables in the authors' FEMFFUSION code, and that with those `GARNET` reproduces them too.

The Robin vacuum lowers $k$ by 52–54 pcm for every method except SDP1 (−73 pcm). It was the default of every solver except the dedicated SP3 when this study started; Marshak is now the default wherever it is available, and Robin is kept for comparison.

The angular error, $k_\infty - k_{\mathrm{OpenMOC}}$ with Marshak, is −2856 pcm for diffusion, −632, −256 and −181 pcm for SP3, SP5, SP7, and −375, −163 and −144 pcm for SDP1, SDP2, SDP3. At equal numbers of moments $\mathrm{SDP}_N$ is closer to transport than $\mathrm{SP}_{2N+1}$ (SDP1/SP3, SDP2/SP5, SDP3/SP7), the paper's central claim, and it holds with the re-derived SDP3 tables. The two transport references differ by 77 pcm (TWODANT $S_{16}$ above OpenMOC), which is about half the remaining SDP3 and SP7 errors: the references do not resolve which of those two is closer to transport.

## 3.4 Error budget on the test mesh

Table 3 splits the single-mesh comparison into its parts. With Marshak, every method is +87 to +127 pcm above the paper on the $80 \times 80$ mesh, and all of it is mesh error (for SDP3, less the −10 pcm of its re-derived tables). With Robin the offset is smaller, +45 to +73 pcm, only because two errors of opposite sign add:

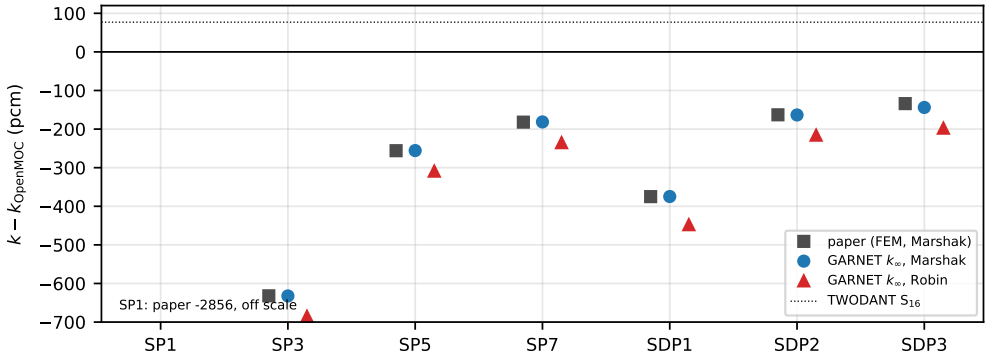



Figure 2: Brantley–Larsen eigenvalues relative to OpenMOC: paper (finite elements, Marshak), `GARNET` mesh limit with Marshak and with Robin vacuum. TWODANT $S_{16}$ is the dotted line.

Table 3: Brantley–Larsen error budget (pcm). Mesh error is $k_{80} - k_\infty$ with Marshak. The last two columns are the single-mesh comparison of the validation test on the $80 \times 80$ mesh, with each vacuum treatment.

| Method | Mesh error at $n = 80$ | Robin − Marshak | Angular ($k_\infty - k_{\mathrm{OpenMOC}}$) | $k_{80}$− paper, Marshak | $k_{80}$− paper, Robin |
|---|---|---|---|---|---|
| SP1 | +86.6 | – | -2856 | +86.7 | – |
| SP3 | +114.1 | -52.9 | -632 | +113.8 | +60.8 |
| SP5 | +121.9 | -53.3 | -256 | +122.3 | +68.8 |
| SP7 | +126.8 | -53.8 | -181 | +127.4 | +73.3 |
| SDP1 | +117.4 | -73.0 | -375 | +117.7 | +44.6 |
| SDP2 | +126.6 | -52.3 | -163 | +126.2 | +73.4 |
| SDP3 | +132.0 | -53.7 | -144 | +122.1 | +67.9 |

mesh error and the Robin offset (−52 to −73 pcm). Under the old Robin defaults the test's 150 pcm envelope therefore passed partly by cancellation; it now checks the Marshak results, and the Robin offset separately.

## 3.5 Equivalent code paths

Table 4: Dedicated solvers against the general $\mathrm{SP}_N/\mathrm{SDP}_N$ block of the same order.

| Comparison | Vacuum | max $\lvert\Delta k\rvert$, $n = 40$–$640$ | Note |
|---|---|---|---|
| SP3 dedicated vs SPN block | Marshak | 2.2e-10 | same condition |
| SP3 dedicated vs SPN block | Robin | 6.8e-06 | per-moment term on different moments |
| SDP1 dedicated vs SDPN block | Robin | 3.5e-06 | per-moment term on different moments |

SP3 and SDP1 each have a dedicated two-moment solver and can also be run through the general block of order 1 (Table 4). With Marshak, the dedicated SP3 solver and the $\mathrm{SP}_N$ block agree to $2 \times 10^{-10}$ on every mesh: two independent implementations of the coupled boundary. With Robin they differ by up to 0.7 pcm (SP3) and 0.35 pcm (SDP1), because the dedicated solvers apply $\alpha = 1/2$ to $(\Phi_1, \phi_2)$ and the general block to its $U$ moments. The per-moment Robin term is an approximation whose result depends on the choice of moments; the Marshak condition is not.

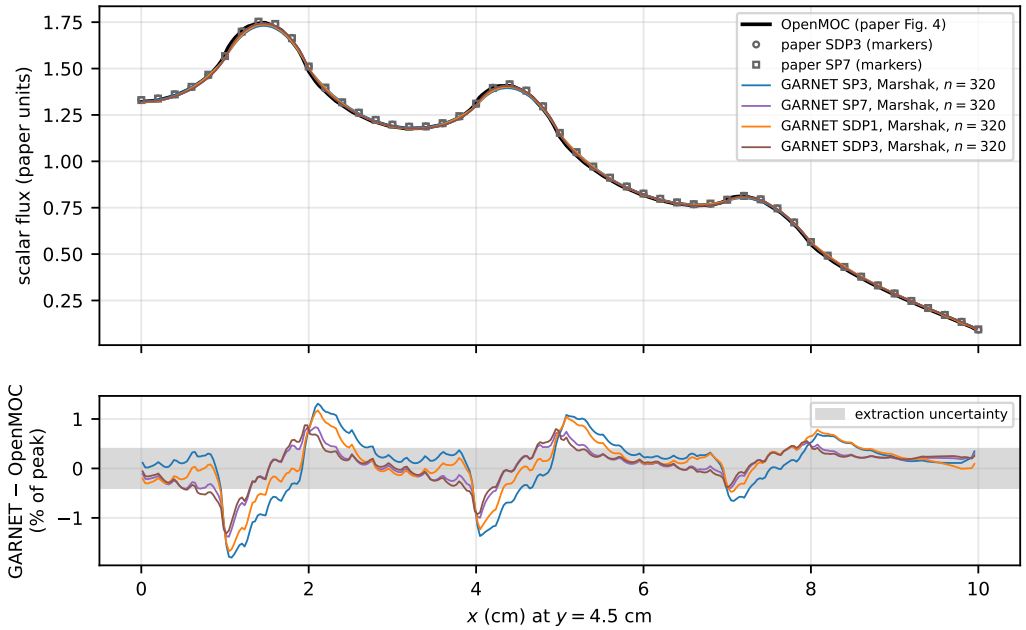



Figure 3: Scalar flux along $y = 4.5$ cm: OpenMOC and the paper's SDP3 and SP7 markers (paper Fig. 4) and `GARNET` with Marshak vacuum. `GARNET` curves are scaled to OpenMOC by least squares. Lower panel: `GARNET` minus OpenMOC; the grey band is the extraction uncertainty.

Table 5: Flux along $y = 4.5$ cm against OpenMOC, after a least-squares scale (% of the OpenMOC peak).

| Solution | RMS (% of peak) | max (% of peak) |
|---|---|---|
| paper SDP3 | 0.39 | 1.40 |
| paper SP7 | 0.40 | 1.39 |
| SP3 | 0.59 | 1.80 |
| SP7 | 0.37 | 1.38 |
| SDP1 | 0.49 | 1.68 |
| SDP3 | 0.36 | 1.32 |

### 3.6 Flux along $y = 4.5$ cm

Fig. 3 and Table 5. `GARNET`'s SDP3 and SP7 deviate from OpenMOC by 0.36–0.37% of the peak (RMS), the paper's own SDP3 and SP7 markers by 0.39–0.40%. Both are at the extraction uncertainty (0.39%), so on this line the two codes agree with transport equally well. The remaining structure, up to 1.8% of the peak, sits at the fuel-bar edges, where the angular flux is most anisotropic; SP3 is the largest there and SDP3 the smallest. The paper's markers, 0.2 cm apart, are too sparse to show it.

### 3.7 The paper's printed SDP3 tables

Three sets of SDP3 coefficient tables are in circulation, and they differ only in the closure rows 2–3:

- **Re-derived** (`GARNET`'s default): rows 2–3 obtained symbolically from the paper's own closure, Eq. (34). The same derivation reproduces the paper's SDP1 and SDP2 tables and all three Marshak matrices exactly.
- **FEMFFUSION**: the tables in the authors' code (`src/matrix_operators/matrix_operators_sdpn.cc`).

- **Paper (A.6)**: the tables as printed. They equal FEMFFUSION's except row 2 of $c^{(3)}$, printed as $(36/55,\ 54/175)$ where the code has $(9/25,\ -54/175)$.

Table 6: Brantley–Larsen SDP3 with Marshak vacuum and the three coefficient sets. $k_\infty$ is extrapolated from $n = 80, 160, 320$.

| SDP3 tables | $k_{80}$ | $k_{160}$ | $k_{320}$ | $p$ | $k_\infty$ (80–320) | − paper |
|---|---|---|---|---|---|---|
| re-derived (`GARNET`) | 0.805241 | 0.804266 | 0.804011 | 1.93 | 0.803921 | -9.9 |
| FEMFFUSION code | 0.805352 | 0.804369 | 0.804111 | 1.93 | 0.804018 | -0.2 |
| paper Eq. (A.6) as printed | 0.805598 | 0.804609 | 0.804349 | 1.93 | 0.804256 | +23.6 |

On Brantley–Larsen (Table 6) the FEMFFUSION tables give a mesh limit 0.2 pcm from the paper's 0.80402, the re-derived tables −9.9 pcm and the printed tables +23.6 pcm. The paper's SDP3 results were therefore computed with the FEMFFUSION tables, and with them `GARNET` reproduces Table 6 for all seven methods. On C5G7 at $s = 4$ the FEMFFUSION tables raise SDP3 by 14.3 pcm and the printed ones by 15.2 pcm.

Which tables are right is a question of derivation, which this benchmark cannot settle. The re-derived rows are the only ones of the three that follow from the closure the paper states, and `tests/verification/test_sdpn_derivation.py` keeps them in place. The choice moves $k$ by 10–15 pcm on these problems, small against SDP3's angular error (−144 pcm against OpenMOC on Brantley–Larsen).

## 3.8 Cost

Table 7: Brantley–Larsen cost at $n = 160$, best of three solves on one CPU thread (merlin7 `cn000`). Inner iterations are operator applies of the within-group Krylov solver. The paper's CPU times are for its own $80 \times 80$ finite-element mesh and are listed only for scale.

| Method | Vacuum | Krylov | outers | inners | solve (s) | paper CPU (s) |
|---|---|---|---|---|---|---|
| SP1 | – | CG | 18 | 3357 | 0.77 | 1.1 |
| SP3 | Marshak | CG | 17 | 3121 | 1.75 | 1.5 |
| SP3 | Robin | CG | 18 | 3224 | 1.52 | 1.5 |
| SP5 | Marshak | CG | 18 | 3139 | 2.66 | 2.0 |
| SP5 | Robin | CG | 18 | 3144 | 2.37 | 2.0 |
| SP7 | Marshak | CG | 18 | 3088 | 3.79 | 2.6 |
| SP7 | Robin | CG | 18 | 3094 | 3.39 | 2.6 |
| SDP1 | Marshak | GMRES | 24 | 4877 | 8.03 | 1.5 |
| SDP1 | Robin | CG | 19 | 3313 | 1.63 | 1.5 |
| SDP2 | Marshak | GMRES | 26 | 4738 | 12.95 | 2.0 |
| SDP2 | Robin | CG | 18 | 3081 | 2.25 | 2.0 |
| SDP3 | Marshak | GMRES | 29 | 4923 | 18.71 | 2.6 |
| SDP3 | Robin | GMRES | 28 | 4803 | 17.58 | 2.6 |

Table 7. At $n = 160$ (25 600 cells, one group) the $\mathrm{SP}_N$ solves take 0.8–3.8 s with nearly the same inner work for every order (3088–3357 operator applies): the cost grows with the size of the block, not with the number of iterations, and the Marshak boundary costs $\mathrm{SP}_N$ nothing measurable (CG either way). SDP1 and SDP2 with Robin vacuum are symmetrized and solved by CG at about the cost of SP3 and SP5. With Marshak their block is non-symmetric and runs GMRES: 1.5 times the applies but 5–6 times the time (8.0 and 13.0 s), the rest being GMRES's orthogonalization. SDP3 has no symmetric form once a vacuum face is present and runs GMRES with either boundary (18–19 s). The paper's CPU times refer to its own mesh and machine and are listed for scale only.

# 4 C5G7

## 4.1 Models and runs

The quarter core is $51 \times 51$ pin cells of 1.26 cm (four $17 \times 17$ assemblies and a 21.42 cm water reflector), seven groups, reflective on the two symmetry planes and vacuum outside. Three `GARNET` models were run:

- **Pin-resolved** (the paper's geometry): every lattice position is an $r = 0.54$ cm cylinder in water, on $s \times s$ cells per pin cell, with exact covered fractions (`build_c5g7_2d(pin_resolved=True, exact_area=True, outer_bc="vacuum")`). With Marshak vacuum, $s = 4$–$48$ for $\mathrm{SP}_N$ and $s = 4$–$24$ for $\mathrm{SDP}_N$; Robin vacuum at $s = 4, 8, 16$; zero-flux outer faces at $s = 8$.
- **Pin-resolved with sampled areas**: the builder's former default, $16 \times 16$-point sampling of the covered fractions (`exact_area=False`), SP1 only.
- **Pin-cell homogenized**: each pin cell volume-weighted, zero-flux outer faces, $s = 1, 2, 4, 8$. This is the model of `tests/validation/test_c5g7.py` (at $s = 1$).

Up to $s = 12$ runs were made on the CPU, from $s = 8$ on a GH200 (`data/runs-gh200/`). The 18 configurations run on both agree to $1.3 \times 10^{-13}$ in $k$.

## 4.2 Mesh convergence

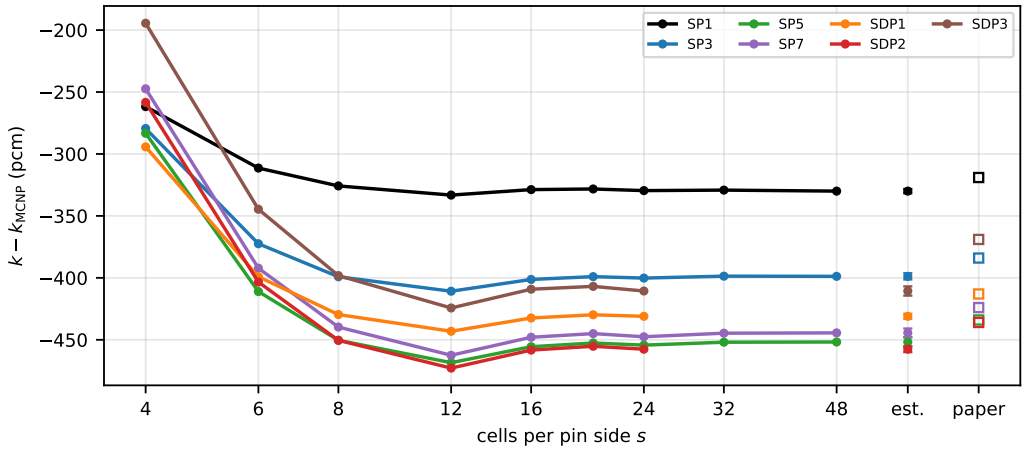



Figure 4: C5G7 pin-resolved, Marshak vacuum: $k - k_{\mathrm{MCNP}}$ against cells per pin side. "est." is the finest-mesh estimate with its spread over $s \geq 16$; "paper" is Table 7.

The pin-resolved model does not converge monotonically (Fig. 4, Table 8). At $s = 4$ every method is 68–216 pcm above its fine-mesh value; at $s = 12$ every method dips 3–18 pcm below it; from $s = 16$ on the values settle. How the fuel circle cuts the cells changes from mesh to mesh, so the blend error does not follow a power law, and Richardson extrapolation through these points would be arbitrary. The estimate is therefore the finest-mesh $k$ ($s = 48$ for $\mathrm{SP}_N$, 24 for $\mathrm{SDP}_N$). Its uncertainty, 1.7–3.8 pcm, is the larger of two measures: the largest difference from it among the meshes with $s \geq 16$, and its distance to a least-squares $k_\infty + b/s^2$ fit over the same meshes.

## 4.3 Comparison with the paper and MCNP

For the six methods whose tables the paper and `GARNET` share (SP1–SP7, SDP1, SDP2), `GARNET`'s estimate is 11–22 pcm below Table 7, and the offset grows with the number of moments: 11 pcm for diffusion, 15–20 pcm for SP3–SP7, 18–22 pcm for SDP1–SDP2. For SDP3 it is 42 pcm. The FEMFFUSION tables

Table 8: C5G7 pin-resolved eigenvalues, Marshak vacuum. Paper column: Table 7 $k$ and $10^5(k - k_{\mathrm{MCNP}})$. Differences in pcm.

| Method | Paper (pcm) | $s = 4$ | $s = 8$ | $s = 12$ | $s = 16$ | $s = 24$ | $s = 32$ | $s = 48$ | $k$ estimate | ± (pcm) | − MCNP | − paper |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| SP1 | 1.18336 (-319) | 1.18393 | 1.18329 | 1.18322 | 1.18326 | 1.18325 | 1.18326 | 1.18325 | 1.18325 | 1.7 | -330 | -11 |
| SP3 | 1.18271 (-384) | 1.18376 | 1.18256 | 1.18244 | 1.18254 | 1.18255 | 1.18256 | 1.18256 | 1.18256 | 2.5 | -399 | -15 |
| SP5 | 1.18221 (-434) | 1.18372 | 1.18205 | 1.18187 | 1.18199 | 1.18201 | 1.18203 | 1.18203 | 1.18203 | 3.8 | -452 | -18 |
| SP7 | 1.18231 (-424) | 1.18408 | 1.18215 | 1.18193 | 1.18207 | 1.18207 | 1.18210 | 1.18211 | 1.18211 | 3.5 | -444 | -20 |
| SDP1 | 1.18242 (-413) | 1.18361 | 1.18225 | 1.18212 | 1.18223 | 1.18224 | – | – | 1.18224 | 2.1 | -431 | -18 |
| SDP2 | 1.18219 (-436) | 1.18397 | 1.18205 | 1.18182 | 1.18197 | 1.18197 | – | – | 1.18197 | 2.4 | -458 | -22 |
| SDP3 | 1.18286 (-369) | 1.18461 | 1.18257 | 1.18231 | 1.18246 | 1.18244 | – | – | 1.18244 | 3.7 | -411 | -42 |

raise C5G7 SDP3 by 14 pcm at $s = 4$ (Section 3.7), so with them the offset would be about 27 pcm, in line with the other methods.

Unlike Brantley–Larsen, the two C5G7 models are not the same discrete problem. The paper meshes each pin with second-degree finite elements (its Fig. 5) and states neither how the circular pin boundary is represented nor a mesh study; `GARNET` blends the cells the circle cuts, with exact areas. The source of the 11–22 pcm offset is not identified. It is 3–5% of the 320–460 pcm angular error and changes none of the orderings below.

Against MCNP both codes give the same picture. Diffusion is closest (−330 pcm here, −319 pcm in the paper), and every $\mathrm{SP}_N$ and $\mathrm{SDP}_N$ order is further away (−399 to −458 pcm here, −369 to −436 pcm in the paper). This is the stagnation the paper reports for C5G7: on this lightly heterogeneous lattice, the eigenvalue is not a measure of angular accuracy. The pin powers are.

## 4.4 Modelling choices

Table 9: C5G7 at $s = 8$: effect of the vacuum treatment and of pin-cell homogenization on $k$ (pcm).

| Method | Robin − Marshak | zero flux − Marshak | homogenized − pin-resolved (zero flux) |
|---|---|---|---|
| SP1 | – | -0.23 | +379 |
| SP3 | -0.20 | -1.33 | +591 |
| SP5 | -0.28 | -1.35 | +654 |
| SP7 | -0.29 | -1.36 | +652 |
| SDP1 | -0.18 | -1.25 | +626 |
| SDP2 | -0.28 | -1.36 | +662 |
| SDP3 | -0.29 | -1.37 | +618 |

Table 9. The vacuum faces sit behind a full assembly width of water, so the boundary treatment hardly matters here: Robin is 0.2–0.3 pcm below Marshak, and the builder's default zero-flux faces are 0.2–1.4 pcm below. The pin-area computation matters much more. With the default sampled fractions, SP1 is 226 pcm higher at $s = 4$ and 22 pcm lower at $s = 8$ than with exact fractions, the result of a fuel-area error of −0.66% and +0.06% (finding 4 explains why sampling errs this way). Pin-cell homogenization raises $k$ by 379 pcm (SP1) to 662 pcm (SDP2) against the pin-resolved model at the same mesh.

## 4.5 Pin powers

Table 10 and Fig. 5. Unlike $k$, the pin powers improve with angular order: AVG falls from 1.43% for diffusion to 0.74% for SP3, 0.66% for SDP1 and 0.53–0.57% for SP5, SP7, SDP2 and SDP3. These are the paper's Table 7 values within 0.04 percentage points for SP1 to SP7, SDP1 and SDP2 (SP1 1.43 against 1.43, SP3 0.74 against 0.75), and the paper's RMSE measure agrees as well. The agreement shows that the

Table 10: C5G7 pin powers against the NEA MCNP reference, Marshak vacuum, finest mesh. Errors in %; the paper's Table 7 AVG and RMSE in brackets. AVG, RMSE as the paper's tables use it ($\sqrt{\sum e_i^2/N}$), RMS, MRE and the largest pin error, the largest pin power (MCNP 2.498), and the assembly power errors.

| Method | $s$ | AVG (paper) | RMSE (paper) | RMS | MRE | max $\lvert e\rvert$ | $P_{\max}$ | inner $UO_2$ | MOX | outer $UO_2$ |
|---|---|---|---|---|---|---|---|---|---|---|
| SP1 | 48 | 1.43 (1.43) | 0.053 (0.05) | 1.72 | 1.22 | 5.79 | 2.522 | +0.57 | -0.45 | -0.65 |
| SP3 | 48 | 0.74 (0.75) | 0.027 (0.03) | 0.88 | 0.67 | 2.81 | 2.516 | +0.46 | -0.52 | -0.06 |
| SP5 | 48 | 0.57 (0.59) | 0.021 (0.02) | 0.69 | 0.51 | 2.93 | 2.500 | +0.13 | -0.16 | +0.03 |
| SP7 | 48 | 0.54 (0.57) | 0.021 (0.02) | 0.67 | 0.49 | 2.88 | 2.491 | -0.10 | +0.10 | +0.05 |
| SDP1 | 24 | 0.66 (0.68) | 0.024 (0.03) | 0.79 | 0.60 | 2.76 | 2.512 | +0.38 | -0.43 | -0.02 |
| SDP2 | 24 | 0.53 (0.57) | 0.020 (0.02) | 0.65 | 0.49 | 2.75 | 2.492 | -0.06 | +0.05 | +0.03 |
| SDP3 | 24 | 0.56 (0.64) | 0.022 (0.03) | 0.72 | 0.52 | 2.65 | 2.481 | -0.34 | +0.38 | +0.06 |

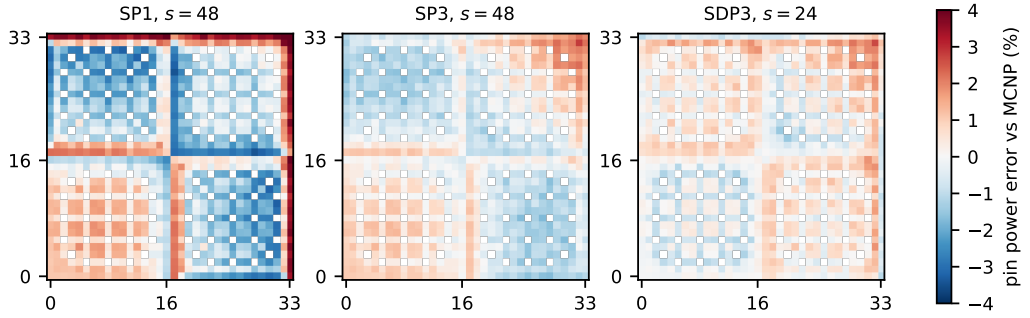



Figure 5: Pin power error against MCNP in the four fuel assemblies (inner $UO_2$ at the bottom left, MOX on the off-diagonal, outer $UO_2$ at the top right).

paper scored its pin powers against the NEA MCNP reference used here, and that the two codes' solutions match pin by pin, not only in $k$.

For the higher orders AVG converges with mesh more slowly than $k$ and from below: SP7 rises from 0.47% at $s = 8$ to 0.54% at $s = 48$, SDP3 from 0.49% at $s = 8$ to 0.56% at $s = 24$, still climbing. The remaining gap to the paper for SP7, SDP2 and SDP3 (0.03–0.08 points, largest for SDP3 at its coarser finest mesh) is consistent with that. Pin-power results from $s \leq 12$ therefore understate the converged error.

The largest pin error is 2.6–2.9% for every transport method and 5.8% for diffusion. In every case it is a corner pin of the outer $UO_2$ assembly next to the reflector, where the power is lowest (0.23–0.29) and the MCNP reference itself is least precise. The largest pin power is within 1% of MCNP's 2.498, and every assembly power within 0.65%.

## 4.6 Flux along the assembly diagonals

Fig. 6 and Table 11. `GARNET`'s SDP3 deviates from OpenMC by 1.1–1.7% of the peak (RMS) on the four panels, the paper's SDP3 by 1.1–2.0%: the two agree with the Monte Carlo profile equally well, and `GARNET`'s SDP3 is closer in group 7 of the MOX diagonal (1.7 against 2.0%). `GARNET`'s SP3 is further away, 1.5–1.9%, as in the paper's own $SP_N/SDP_N$ comparison. The largest local deviations, up to 8% of the peak in group 7, sit on the thermal-flux peaks in the water between pins and at the reflector-side corner of the MOX assembly. The OpenMC curve is steepest there, so these are also the points where the extracted reference is least precise.

## 4.7 The homogenized model of the regression test

The regression test's model, pin-cell homogenized at $s = 1$, lands 8–165 pcm below MCNP (Table 12). The agreement is a cancellation of two large errors. Homogenizing each pin cell raises $k$ by 380–660 pcm

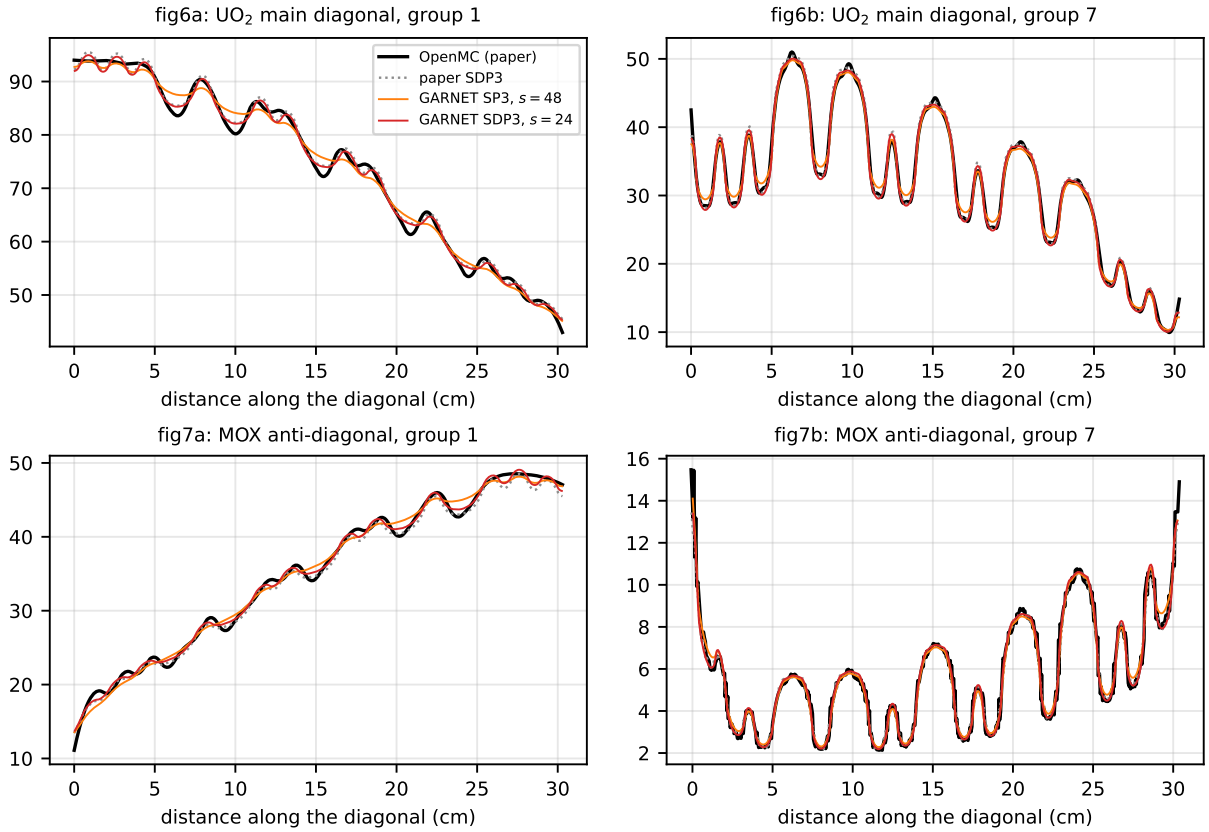



Figure 6: Groups 1 and 7 along the inner $UO_2$ main diagonal (paper Fig. 6) and the MOX anti-diagonal (Fig. 7): OpenMC and the paper's SDP3, both read from the paper, and `GARNET` SP3 and SDP3 scaled to OpenMC by least squares.

against the pin-resolved model (Table 9), so the homogenized mesh limit is 57–223 pcm *above* MCNP and 376–650 pcm above the paper; the $s = 1$ mesh then lowers $k$ by 222–231 pcm. The test's eigenvalue envelopes (2% for diffusion, 0.5% for the family) are therefore not accuracy statements. What those tests guard is the internal ordering of the hierarchy and the clustering of the six transport methods, which is where the paper's SDP2/SDP3 table errors showed.

## 4.8 Solver behaviour and cost

Table 13. On one CPU thread at $s = 4$ (41 616 cells, seven groups) the $SP_N$ solves take 6–40 s and the $SDP_N$ solves 49–124 s. With Marshak vacuum every $SDP_N$ block is non-symmetric and runs GMRES. At $s = 4$ its outer iteration count is still close to $SP_N$'s (64–106 against 59–68), but from $s = 8$ on it roughly doubles (121–161 against 57–73), and at $s = 24$ the inner applies are 2.9–3.6 times those of SP3–SP7. On the GH200 at $s = 24$ (1.5 million cells) $SP_N$ takes 30–271 s and $SDP_N$ 1580–1969 s. At $s = 32$ none of the $SDP_N$ solves converged: in the first outer iteration GMRES with Jacobi preconditioning stopped at 5000 iterations, 1.3–4.8 times above its tolerance (finding 5 in Section 7). The $SP_N$ solves ran to $s = 48$ (6.0 million cells) in 2–14 minutes each.

The pin-resolved study is a GPU calculation: on the shared CPU node, with up to 16 runs at once, the $SP_N$ solves at $s = 12$ took 14–77 minutes each and an SDP3 solve at $s = 12$ was projected at over 20 hours.

Table 11: Flux along the diagonals against OpenMC, after a least-squares scale (% of the OpenMC peak).

| Figure | Solution | RMS (% of peak) | max (% of peak) |
|---|---|---|---|
| fig6a | paper SDP3 | 1.10 | 2.49 |
| fig6a | GARNET SP3 | 1.72 | 4.18 |
| fig6a | GARNET SDP3 | 1.08 | 2.32 |
| fig6b | paper SDP3 | 1.36 | 8.15 |
| fig6b | GARNET SP3 | 1.49 | 8.49 |
| fig6b | GARNET SDP3 | 1.21 | 5.64 |
| fig7a | paper SDP3 | 1.22 | 5.23 |
| fig7a | GARNET SP3 | 1.86 | 4.66 |
| fig7a | GARNET SDP3 | 1.22 | 4.72 |
| fig7b | paper SDP3 | 2.04 | 15.22 |
| fig7b | GARNET SP3 | 1.90 | 10.86 |
| fig7b | GARNET SDP3 | 1.70 | 8.31 |

Table 12: C5G7 pin-cell homogenized, zero-flux outer faces. $k_\infty$ from $s = 4, 8$ at second order. Differences in pcm.

| Method | $s = 1$ (test) | $s = 2$ | $s = 4$ | $s = 8$ | $k_\infty$ | test − MCNP | $k_\infty$− MCNP | $k_\infty$− paper |
|---|---|---|---|---|---|---|---|---|
| SP1 | 1.18490 | 1.18650 | 1.18696 | 1.18708 | 1.18712 | -165 | +57 | +376 |
| SP3 | 1.18611 | 1.18779 | 1.18830 | 1.18846 | 1.18851 | -44 | +196 | +580 |
| SP5 | 1.18630 | 1.18795 | 1.18843 | 1.18858 | 1.18863 | -25 | +208 | +642 |
| SP7 | 1.18640 | 1.18806 | 1.18852 | 1.18866 | 1.18870 | -15 | +215 | +639 |
| SDP1 | 1.18617 | 1.18784 | 1.18834 | 1.18850 | 1.18855 | -38 | +200 | +613 |
| SDP2 | 1.18639 | 1.18805 | 1.18851 | 1.18865 | 1.18869 | -16 | +214 | +650 |
| SDP3 | 1.18647 | 1.18816 | 1.18861 | 1.18873 | 1.18878 | -8 | +223 | +592 |

Table 13: C5G7 cost, Marshak vacuum. CPU: best of three at $s = 4$, one thread. GH200: single runs, which include CuPy kernel compilation on first use; outer and inner counts at $s = 24$.

| Method | Krylov | outers | inners | CPU $s = 4$ (s) | GH200 $s = 8$ (s) | outers $s = 24$ | inners $s = 24$ | GH200 $s = 24$ (s) |
|---|---|---|---|---|---|---|---|---|
| SP1 | CG | 61 | 17347 | 6.0 | 9.2 | 60 | 129535 | 30.5 |
| SP3 | CG | 62 | 18641 | 15.4 | 61.8 | 73 | 161131 | 245.1 |
| SP5 | CG | 68 | 20778 | 29.1 | 69.8 | 64 | 141625 | 232.5 |
| SP7 | CG | 59 | 17579 | 40.0 | 76.7 | 67 | 143091 | 270.6 |
| SDP1 | GMRES | 64 | 17242 | 49.0 | 223.8 | 144 | 509771 | 1590.2 |
| SDP2 | GMRES | 90 | 20432 | 84.2 | 157.6 | 141 | 462018 | 1579.7 |
| SDP3 | GMRES | 106 | 22254 | 124.3 | 191.9 | 158 | 464549 | 1968.7 |

# 5 GPU and CPU performance

**Interim version, 2 November 2026.** Two single-core runs of the extruded cases (Brantley–Larsen SDP3 with 16 layers, C5G7 SDP3 with 8 layers) and the two concurrency controls of the extruded jobs are still running and are expected to finish by the evening of 3 November. This section quotes projections for them, and for the extruded GMRES cases that will not be measured on one core; it will be updated with the measurements.

This section compares `GARNET` on one NVIDIA GH200 with `GARNET` on one core of a CPU node, on identical meshes, for SP1 (diffusion, CG), SP3 (CG), SDP1 and SDP3 (both GMRES), all with the Marshak vacuum and the tolerances of the accuracy study. It answers a different question from the rest of the report, so the meshes go far past convergence, towards the cell counts of a 3-D core. FEMFFUSION was not run; the paper's own FEMFFUSION times for Brantley–Larsen (Table 6: 1.1–2.6 s on an $80 \times 80$ second-degree mesh, hardware not stated) are quoted for scale only.

**Two kinds of mesh.** The 2-D ladders refine each problem until the solver stops converging. With Jacobi preconditioning the inner iteration count grows with the 2-D mesh size, and the Krylov solvers stop at 5000 iterations: CG fails at Brantley–Larsen $n = 2560$ (6.6 million cells), GMRES at $n = 1280$ for SDP1 and at C5G7 $s = 32$ (audit item O14). A 3-D core reaches tens of millions of cells with far fewer cells per direction, so it is much better conditioned. The extruded cases reproduce that: the 2-D mesh is repeated over $n_z$ layers of cubic cells with reflective top and bottom, so the solution, and $k$, are exactly those of the 2-D problem while the grid, its memory traffic and its arithmetic are genuinely 3-D. Brantley–Larsen $n = 640$ is extruded to 16, 64 and 256 layers (up to 105 million cells), C5G7 $s = 8$ to 8, 64 and 128 layers and $s = 16$ to 64 layers (up to 43 million cells, 7 groups). The largest $\mathrm{SDP}_N$ cases stop at the GH200's memory. On the CPU, the extruded cases go to 64 layers, except SDP3, which goes to 32: at 64 layers it would run past the 5-day job limit. The GH200 also runs SDP3 at 32 layers, so that the two devices are compared on the same mesh.

**Timing.** Solve time is measured between device synchronizations. GPU runs keep the best of three solves (the first includes CuPy's kernel compilation), except the largest extruded ones, which are single solves. CPU runs keep the best of three where a solve takes minutes and use a single solve where it takes hours. Each CPU job had an exclusive merlin7 node (two 64-core AMD EPYC 7742) and ran its methods side by side as single-threaded processes, at most five at a time. One case of each job was re-timed alone at the end as a control. `GARNET`'s CPU path is single-threaded, so the speedup is over one core. A code that used every core of the node would narrow it, by up to the node's memory-bandwidth limit.

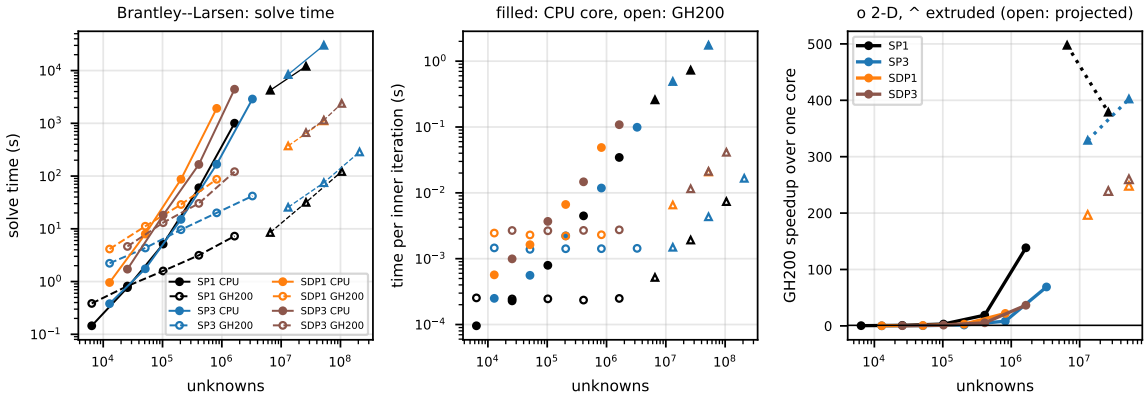



Figure 7: Brantley–Larsen, Marshak vacuum, on identical meshes: solve time (left), time per inner iteration (middle) and the GH200's speedup over one CPU core (right). CPU filled markers and solid lines, GH200 open markers and dashed lines; circles 2-D meshes, triangles extruded ones. In the right panel, open triangles are speedups with a projected one-core time (Table 16).

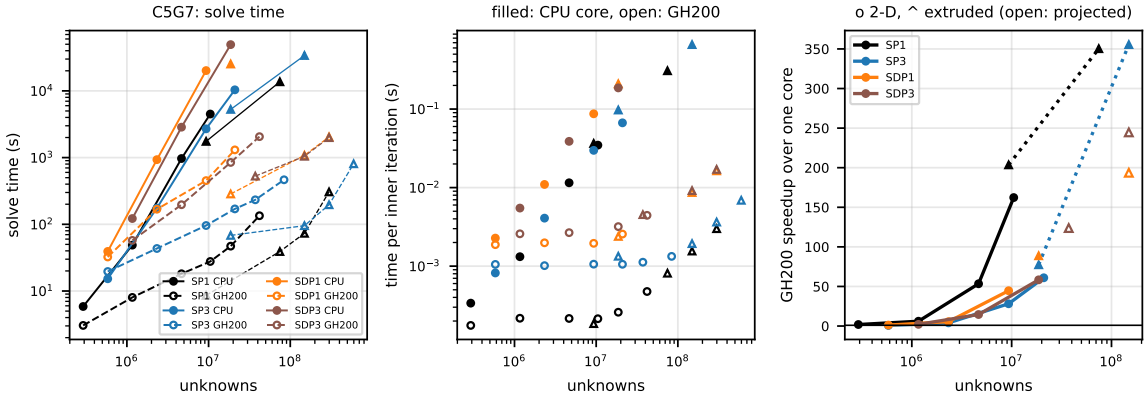



Figure 8: C5G7 pin-resolved, Marshak vacuum: as Fig. 7.

## 5.1 Results

**Same answer on both devices.** Wherever both devices converged, they give the same $k$ within $3\times10^{-10}$ on Brantley–Larsen. On C5G7 the two runs can stop a few outer iterations apart and differ by up to $7\times10^{-9}$, inside the $2 \times 10^{-8}$ allowed between equivalent configurations. The extruded cases give the $k$ of the 2-D mesh they repeat within $5 \times 10^{-9}$.

**Crossover.** On Brantley–Larsen the GH200 is faster than one core from 0.1–0.2 million unknowns: $n = 160$ for SDP3 and $n = 320$ for the other methods, where the CPU solve takes 5–90 s. On C5G7 it is faster from the coarsest mesh, $s = 4$ (0.3–1.2 million unknowns), for every method except SP3, which crosses at $s = 8$. Below the crossover the GH200 is up to six times slower, because its fixed cost per iteration (below) dominates.

**Speedup on the largest meshes both devices ran** (Tables 14 and 15):

| | SP1 | SP3 | SDP1 | SDP3 |
|---|---|---|---|---|
| Brantley–Larsen | 138 ($n = 1280$) | 69 ($n = 1280$) | 22 ($n = 640$) | 37 ($n = 640$) |
| C5G7 | 162 ($s = 24$) | 61 ($s = 24$) | 45 ($s = 16$) | 58 ($s = 16$) |

On these meshes one core takes 17 minutes (Brantley–Larsen SP1) to 13.7 hours (C5G7 SDP3), and the GH200 7 s to 14 minutes. On every ladder the speedup is still rising at the last mesh.

**Why the speedup grows** (middle panels of Figs. 7 and 8). The cost of one inner iteration behaves differently on the two devices.

- *GH200.* The time per inner iteration is constant up to a few million cell-moments: 0.18–0.25 ms for SP1, 1.0–1.4 ms for SP3, 1.9–2.6 ms for SDP1 and 2.6–3.2 ms for SDP3. In this regime the solve time depends only on the iteration count, not on the mesh. Above it the time grows linearly with the mesh. On the largest meshes it is 0.07–0.08 ns per cell-moment per iteration for the CG methods (SP1, SP3) and 0.38–0.41 ns for the GMRES methods (SDP1, SDP3), whose iterations also orthogonalize against the stored Krylov basis.
- *CPU core.* The time grows linearly from the start, at 8–12 ns per cell-moment per iteration for SP1 and SP3 while a group's vectors fit in cache. It rises two- to threefold, to 21–30 ns, on the largest 2-D meshes; for SDP1 and SDP3 it goes from 27–36 ns to 60–70 ns.

Once both devices are in their linear regimes, the speedup tends to the ratio of these per-unknown costs. The 2-D costs give about 280–380 for the CG methods and 150–180 for GMRES. On the extruded meshes one core is slower per unknown than on the largest 2-D meshes, so the speedups there come out higher: 350–400 measured for CG and about 200–260 projected for GMRES (below). Part of the largest speedups comes from the CPU working set leaving the cache, which is what a realistic 3-D problem does on a CPU too.

**SP3 has a high fixed cost on the GH200.** SP3's time per iteration at small sizes (1.0–1.4 ms) is five to six times SP1's, although its block has only twice the unknowns. On the largest meshes the ratio falls to 2.2 (BL with 256 layers: 16.6 against 7.4 ms per iteration), as the extra unknowns predict. The extra fixed cost is per operator apply. The likely source is the coupled Marshak term: `_apply_marshak` in `garnet/spn.py` applies it as gathers, a batched $2 \times 2$ product and scatters over the boundary cells, each a separate small kernel. This has not been profiled. If SP3's fixed cost were twice SP1's, mid-sized SP3 solves on the GPU would be up to about three times faster. It does not affect the CPU or the results.

**3-D-sized problems on one GH200.** The extruded cases reach the cell counts of a 3-D core. They need 15–58% more inner iterations than the 2-D mesh they repeat, and the number does not grow with the layer count. A 2-D ladder, by contrast, doubles its inner iterations with each mesh doubling. The largest cases solved:

- Brantley–Larsen with 256 layers (105 million cells): SP1 in 122 s and SP3 (210 million unknowns) in 287 s.
- C5G7 at $s = 16$ with 64 layers (42.6 million cells, 7 groups): SP1 (298 million unknowns) in 306 s and SP3 (597 million) in 807 s, using 77 GB.
- C5G7 at $s = 8$: SDP1 with 128 layers and SDP3 with 64 layers (298 million unknowns each) in 2040 and 1990 s.

After the solve, CuPy's memory pool held 130–250 bytes per unknown on the extruded C5G7 cases (SP3 lowest, about 210–220 for SDP1 and SDP3), and 260–510 on the one-group Brantley–Larsen problem, where per-cell data is shared by fewer unknowns. Two cases ran out of the GH200's 96 GiB (102.6 GB), both at 597 million unknowns: SDP1 at $s = 16$ with 64 layers and SDP3 at $s = 8$ with 128. Both stopped in the first GMRES solve with 100.6 GB allocated, of which the 61-vector Krylov basis alone is 42 GB. By these figures, one GH200 holds a 7-group 3-D problem of roughly 450 million unknowns with SDP1 or SDP3, and 400–800 million with SP1 or SP3.

**Extruded cases on one core.** The CPU runs covered all four methods at 16 and 64 layers of Brantley–Larsen and at 8 and 64 layers of C5G7 at $s = 8$, with SDP3 at 32 layers instead of 64: at 64 layers it would take 110–140 h on one core, beyond the 5-day job limit, so the GH200 also ran SDP3 at 32 layers for the comparison.

- *Measured:* SP1 and SP3 at every extruded size, and C5G7 SDP1 at 8 layers. With 64 layers the GH200 is 350–400 times faster than one core: 379 (SP1) and 402 (SP3) on Brantley–Larsen, 350 and 355 on C5G7. Where the GH200 is still in its fixed-cost regime the speedup is lower: 77 for C5G7 SP3 and 89 for C5G7 SDP1, both at 8 layers. On the extruded meshes one core takes 28–37 ns per cell-moment per iteration for SP1 and SP3, somewhat more than on the largest 2-D meshes, and 79 ns for C5G7 SDP1. Brantley–Larsen SP1 at 16 layers took 39 ns, 40% more than at 64 layers, which gives it a speedup of 498. Whether contention with the other runs inflated it will show in the alone re-run at the end of the job.
- *Failed at the iteration cap:* Brantley–Larsen SDP1 at 16 layers and SDP3 at 32 layers, although the GH200 converged on the same meshes. Both stopped in the first outer iteration at the 5000-iteration cap, with residuals 1.45 and 1.6 times the target. The CPU orthogonalizes the Krylov basis by modified Gram–Schmidt and the GH200 by two-pass classical Gram–Schmidt. At these sizes the Jacobi-preconditioned GMRES is so close to the cap that this rounding difference decides the outcome (finding 5).
- *Stopped on purpose:* Brantley–Larsen SDP1 at 64 layers and C5G7 SDP1 at 64 layers and SDP3 at 32 layers, after 4–11 h of an expected 60–80 h each. They would have added more GMRES points of the kind the projections below already estimate.
- *Still running* (2 November): Brantley–Larsen SDP3 at 16 layers and C5G7 SDP3 at 8 layers.

**Projected GMRES speedups** (Table 16, open triangles in the right panels of Figs. 7 and 8). For an extruded GMRES case without a converged CPU run, the one-core time is projected from three numbers:

- one core's measured time per inner iteration per cell-moment in a basis run;

- the case's number of cell-moments;
- the GH200's inner iteration count for the case.

The two cap runs give the basis at large sizes: their 5000 iterations, timed from their logs, took 98 ns (SDP1, 16 layers) and 105 ns (SDP3, 32 layers) per cell-moment. C5G7 SDP1 at 8 layers (79 ns, converged) gives the basis for C5G7 SDP1. C5G7 SDP3 has no measurement yet and borrows the Brantley–Larsen SDP3 cost. The projected speedups are 194–260 at 13–149 million unknowns, and 124 for C5G7 SDP3 at 8 layers, where the GH200 is still near its fixed-cost regime. Three sources of uncertainty add up to about ±25%:

- the CPU and GH200 inner iteration counts of converged runs differ by up to 7%;
- the cap runs' iterations are full restart cycles of a first outer iteration, which may cost slightly more per iteration than the average;
- a cost carried over from another mesh or problem adds the spread seen between measured per-cell costs (79–105 ns for GMRES).

At 3-D sizes the GH200 is therefore about 200–260 times faster than one core for SDP1 and SDP3 (projected), against 350–400 times for SP1 and SP3 (measured): GMRES's orthogonalization takes a larger share of the GH200's time than of the CPU's.

**Concurrency control.** Re-timed alone after the concurrent runs had finished, Brantley–Larsen SP3 at $n = 640$ took 166.3 s against 167.3 s side by side (0.6%), and C5G7 SP3 at $s = 8$ took 169.0 s against 174.8 s (3.4%). Running the methods side by side inflates the 2-D CPU times by a few percent at most, and the speedups by the same amount. The extruded jobs each end with a similar control (Brantley–Larsen SP1 at 16 layers and C5G7 SP1 at 8 layers, re-timed alone), still pending on 2 November.

**FEMFFUSION, for scale.** The paper reports 1.1–2.6 s for Brantley–Larsen with FEMFFUSION, using second-degree elements on an $80 \times 80$ mesh (about 26 000 nodes per moment) on unstated hardware. Its $k$ values agree with `GARNET`'s mesh limits within 0.6 pcm (Section 3). `GARNET`'s finite volumes reach a comparable mesh error (about 2 pcm) at $n = 640$, where one core takes 60–4400 s and the GH200 3–121 s. At this problem size a higher-order discretization does better than a faster device. The GH200 pays off at the sizes of a 3-D core, where any discretization has tens to hundreds of millions of unknowns.

Table 14: Brantley–Larsen solve times on identical $n \times n$ meshes. CPU: best of three on one core, or a single solve ([1]) where one takes hours. GH200: best of three, single solves for the largest extruded cases. Extruded rows ($n_z > 1$) follow the 2-D ones. Iteration counts are the GH200 run's; GH200 GB is the memory CuPy's pool held after the solve; $|\Delta k|$ is the CPU–GPU difference. OOM: out of GH200 memory; cap: an inner solve hit its 5000-iteration cap; cancelled: stopped on purpose; –: not run. One-core projections for the extruded GMRES cases are in Table 16.

| Method | $n$ | $n_z$ | unknowns ($10^6$) | outers | inners | CPU (s) | GH200 (s) | GH200 GB | speedup | $\lvert\Delta k\rvert$ |
|---|---|---|---|---|---|---|---|---|---|---|
| SP1 | 80 | 1 | 0.0064 | 18 | 1515 | 0.145 | 0.385 | 0.0 | 0.4 | 7e-15 |
| | 160 | 1 | 0.0256 | 18 | 3357 | 0.768 | 0.821 | 0.0 | 0.9 | 3e-14 |
| | 320 | 1 | 0.102 | 17 | 6469 | 5.13 | 1.59 | 0.1 | 3.2 | 1e-13 |
| | 640 | 1 | 0.41 | 17 | 13401 | 59.9 | 3.15 | 0.2 | 19 | 6e-13 |
| | 1280 | 1 | 1.64 | 18 | 29155 | 1005[1] | 7.26 | 0.9 | 138 | 2e-12 |
| | 2560 | 1 | 6.55 | – | – | cap | cap | – | – | – |
| | 640 | 16 | 6.55 | 17 | 16393 | 4235[1] | 8.51 | 3.5 | 498 | 8e-13 |
| | 640 | 64 | 26.2 | 17 | 16526 | 12019[1] | 31.7 | 11.1 | 379 | 8e-13 |
| | 640 | 256 | 105 | 17 | 16566 | – | 122 | 44.5 | – | – |
| SP3 | 80 | 1 | 0.0128 | 18 | 1523 | 0.38 | 2.22 | 0.0 | 0.2 | 7e-15 |
| | 160 | 1 | 0.0512 | 17 | 3121 | 1.75 | 4.33 | 0.0 | 0.4 | 3e-14 |
| | 320 | 1 | 0.205 | 18 | 6782 | 15.1 | 9.61 | 0.1 | 1.6 | 1e-13 |
| | 640 | 1 | 0.819 | 18 | 14140 | 167 | 20.1 | 0.3 | 8.3 | 6e-13 |
| | 1280 | 1 | 3.28 | 18 | 29141 | 2888[1] | 41.8 | 1.1 | 69 | 2e-12 |
| | 2560 | 1 | 13.1 | – | – | cap | cap | – | – | – |
| | 640 | 16 | 13.1 | 18 | 17155 | 8415[1] | 25.6 | 4.6 | 329 | 8e-13 |
| | 640 | 64 | 52.4 | 18 | 17301 | 30121[1] | 74.9 | 13.4 | 402 | 8e-13 |
| | 640 | 256 | 210 | 18 | 17247 | – | 287 | 53.7 | – | – |
| SDP1 | 80 | 1 | 0.0128 | 21 | 1688 | 0.959 | 4.14 | 0.0 | 0.2 | 8e-15 |
| | 160 | 1 | 0.0512 | 24 | 4877 | 7.96 | 11.2 | 0.0 | 0.7 | 3e-14 |
| | 320 | 1 | 0.205 | 23 | 13064 | 87 | 28.9 | 0.1 | 3.0 | 3e-13 |
| | 640 | 1 | 0.819 | 26 | 37611 | 1920[1] | 86.9 | 0.5 | 22 | 3e-11 |
| | 1280 | 1 | 3.28 | – | – | cap | cap | – | – | – |
| | 2560 | 1 | 13.1 | – | – | – | cap | – | – | – |
| | 640 | 16 | 13.1 | 28 | 56920 | cap | 372 | 8.3 | – | – |
| | 640 | 64 | 52.4 | 27 | 53681 | cancelled | 1113 | 26.8 | – | – |
| SDP3 | 80 | 1 | 0.0256 | 25 | 1728 | 1.72 | 4.63 | 0.0 | 0.4 | 8e-15 |
| | 160 | 1 | 0.102 | 29 | 4923 | 18.1 | 13.1 | 0.1 | 1.4 | 3e-14 |
| | 320 | 1 | 0.41 | 28 | 11284 | 166 | 30.3 | 0.2 | 5.5 | 1e-13 |
| | 640 | 1 | 1.64 | 31 | 44192 | 4431[1] | 121 | 1.0 | 37 | 3e-10 |
| | 1280 | 1 | 6.55 | – | – | cap | cap | – | – | – |
| | 2560 | 1 | 26.2 | – | – | – | cap | – | – | – |
| | 640 | 16 | 26.2 | 32 | 57678 | – | 665 | 16.2 | – | – |
| | 640 | 32 | 52.4 | 30 | 54049 | cap | 1146 | 25.4 | – | – |
| | 640 | 64 | 105 | 32 | 57705 | – | 2384 | 50.7 | – | – |

Table 15: C5G7 pin-resolved solve times on identical meshes ($s$ cells per pin side); as Table 14.

| Method | $s$ | $n_z$ | unknowns ($10^6$) | outers | inners | CPU (s) | GH200 (s) | GH200 GB | speedup | $\lvert\Delta k\rvert$ |
|---|---|---|---|---|---|---|---|---|---|---|
| SP1 | 4 | 1 | 0.291 | 61 | 17347 | 5.86 | 3.05 | 0.1 | 1.9 | 2e-15 |
| | 8 | 1 | 1.17 | 61 | 36964 | 48.7 | 8.02 | 0.4 | 6.1 | 1e-14 |
| | 16 | 1 | 4.66 | 61 | 84128 | 969[1] | 18.1 | 2.0 | 53 | 2e-13 |
| | 24 | 1 | 10.5 | 60 | 129535 | 4499[1] | 27.7 | 4.1 | 162 | 6e-14 |
| | 32 | 1 | 18.6 | 61 | 181662 | – | 47.1 | 7.3 | – | – |
| | 48 | 1 | 41.9 | 62 | 282800 | – | 134 | 15.7 | – | – |
| | 8 | 8 | 9.32 | 60 | 46380 | 1749[1] | 8.59 | 3.7 | 204 | 4e-09 |
| | 8 | 64 | 74.6 | 62 | 48078 | 13687[1] | 39.1 | 18.9 | 350 | 3e-09 |
| | 8 | 128 | 149 | 59 | 47122 | – | 73.3 | 37.9 | – | – |
| | 16 | 64 | 298 | 60 | 102062 | – | 306 | 54.2 | – | – |
| SP3 | 4 | 1 | 0.583 | 62 | 18641 | 15.3 | 19.6 | 0.2 | 0.8 | 2e-15 |
| | 8 | 1 | 2.33 | 64 | 42789 | 175 | 43.4 | 0.6 | 4.0 | 3e-15 |
| | 16 | 1 | 9.32 | 64 | 90526 | 2699[1] | 95.9 | 2.4 | 28 | 3e-15 |
| | 24 | 1 | 21 | 73 | 161131 | 10372[1] | 170 | 6.4 | 61 | 3e-09 |
| | 32 | 1 | 37.3 | 68 | 207605 | – | 233 | 9.7 | – | – |
| | 48 | 1 | 83.9 | 71 | 349233 | – | 465 | 21.9 | – | – |
| | 8 | 8 | 18.6 | 63 | 51058 | 5276[1] | 68.2 | 5.1 | 77 | 7e-09 |
| | 8 | 64 | 149 | 60 | 49308 | 33901[1] | 95.4 | 24.7 | 355 | 5e-09 |
| | 8 | 128 | 298 | 66 | 53930 | – | 196 | 49.3 | – | – |
| | 16 | 64 | 597 | 66 | 117107 | – | 807 | 77.4 | – | – |
| SDP1 | 4 | 1 | 0.583 | 64 | 17242 | 39.2 | 32.4 | 0.2 | 1.2 | 9e-14 |
| | 8 | 1 | 2.33 | 122 | 84939 | 932[1] | 168 | 0.8 | 5.5 | 2e-16 |
| | 16 | 1 | 9.32 | 135 | 231530 | 20154[1] | 452 | 3.1 | 45 | 3e-12 |
| | 24 | 1 | 21 | 144 | 509771 | – | 1305 | 7.7 | – | – |
| | 8 | 8 | 18.6 | 131 | 119191 | 25270[1] | 285 | 6.5 | 89 | 5e-11 |
| | 8 | 64 | 149 | 134 | 124965 | cancelled | 1082 | 32.8 | – | – |
| | 8 | 128 | 298 | 134 | 124700 | – | 2038 | 65.6 | – | – |
| | 16 | 64 | 597 | – | – | – | OOM | – | – | – |
| SDP3 | 4 | 1 | 1.17 | 106 | 22254 | 122 | 57.4 | 0.4 | 2.1 | 7e-16 |
| | 8 | 1 | 4.66 | 136 | 73650 | 2859[1] | 197 | 1.7 | 15 | 2e-13 |
| | 16 | 1 | 18.6 | 160 | 265444 | 49385[1] | 846 | 6.7 | 58 | 1e-12 |
| | 24 | 1 | 41.9 | 158 | 464549 | – | 2061 | 15.2 | – | – |
| | 8 | 8 | 37.3 | 157 | 116501 | – | 527 | 13.0 | – | – |
| | 8 | 32 | 149 | 149 | 114215 | cancelled | 1046 | 31.2 | – | – |
| | 8 | 64 | 298 | 153 | 116701 | – | 1990 | 62.0 | – | – |
| | 8 | 128 | 597 | – | – | – | OOM | – | – | – |

Table 16: Projected one-core times of the extruded GMRES cases without a converged CPU run (status of 2 November 2026). CPU run: what happened to the one-core run (cap: stopped at the 5000-iteration cap; cancelled: stopped on purpose; running). ns: one core's time per inner iteration per cell-moment in the basis run; "(cap)" marks a basis timed from a cap run's log. Projected CPU time = ns × cell-moments × GH200 inners; uncertainty about ±25%.

| Case | Method | mesh | $n_z$ | unknowns ($10^6$) | CPU run | basis | ns | GH200 inners | CPU (h) | GH200 (s) | speedup |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Brantley–Larsen | SDP1 | 640 | 16 | 13.1 | cap | `bl-SDP1-n640-z16` (cap) | 98 | 56920 | 20 | 372 | 197 |
| Brantley–Larsen | SDP1 | 640 | 64 | 52.4 | cancelled | `bl-SDP1-n640-z16` (cap) | 98 | 53681 | 77 | 1113 | 248 |
| Brantley–Larsen | SDP3 | 640 | 16 | 26.2 | running | `bl-SDP3-n640-z32` (cap) | 105 | 57678 | 44 | 665 | 239 |
| Brantley–Larsen | SDP3 | 640 | 32 | 52.4 | cap | `bl-SDP3-n640-z32` (cap) | 105 | 54049 | 83 | 1146 | 260 |
| C5G7 | SDP1 | 8 | 64 | 149 | cancelled | `c5g7-SDP1-s8-z8` | 79 | 124965 | 58 | 1082 | 194 |
| C5G7 | SDP3 | 8 | 8 | 37.3 | running | `bl-SDP3-n640-z32` (cap) | 105 | 116501 | 18 | 527 | 124 |
| C5G7 | SDP3 | 8 | 32 | 149 | cancelled | `bl-SDP3-n640-z32` (cap) | 105 | 114215 | 71 | 1046 | 245 |

# 6 Test tolerances

`tests/validation/test_brantley_larsen_2d.py`. The single-mesh tests now check the Marshak results, the default, on the $80 \times 80$ mesh: each method within 150 pcm of Table 6 (measured +87 to +127 pcm, all mesh error) and a positive offset with a spread below 60 pcm (measured 41 pcm). A new test keeps the per-moment Robin results for comparison: Robin minus Marshak between −80 and −45 pcm (measured −53 to −54 pcm, and −73 pcm for SDP1). Before, the single-mesh tests ran each solver's default, mostly Robin, and passed partly by cancellation (Table 3). The mesh-limit test was the weak one: it checked SP3 and SDP1 only, to 10 pcm. It now checks every method with the Marshak vacuum against Table 6 to 4 pcm, from $n = 80$ and 160 at second order; that two-mesh estimate is within 2.2 pcm of the three-mesh limits of Table 1, and the measured distances from the paper are within 1.5 pcm. SP1, SP3, SP5, SP7 and SDP1 run by default (about 20 s together); SDP2 and SDP3 with the FEMFFUSION tables, which need GMRES, are marked `slow`, as is a test that the re-derived SDP3 stays 4–12 pcm below the paper (measured −7.9 pcm from two meshes). A value near zero there would mean the tables had been changed back to the paper's.

`tests/validation/test_c5g7.py`. The homogenized tests keep their envelopes, with corrected explanations: Section 4 shows that their closeness to MCNP is a cancellation, so they guard the ordering and clustering of the hierarchy, not accuracy. Two tests are new. One checks that the default (exact) pin areas of the C5G7 and C5G7-TD builders reproduce $\pi r^2$ at any resolution, that the old sampling is still available and still −0.66% at $s = 4$, and that the vacuum boundary option is passed through. The other, marked `slow` (about a minute), runs pin-resolved diffusion at $s = 8$ with exact areas and vacuum outside, and checks it against the paper's SP1: $k$ within 10 pcm (measured −6.8 pcm) and the pin-power AVG against the NEA MCNP map within 0.06 percentage points of the paper's 1.43% (measured 1.46%). It skips without a FEMFFUSION checkout, which carries that map.

# 7 Findings and recommendations

1. **Code verification passes.** On Brantley–Larsen, with the paper's Marshak boundary, `GARNET`'s mesh limits equal Table 6 within 0.6 pcm for SP1, SP3, SP5, SP7, SDP1 and SDP2, and within 0.2 pcm for SDP3 run with the authors' FEMFFUSION tables. On C5G7 the pin powers reproduce the paper's AVG and RMSE within 0.04 percentage points.

2. **The paper's SDP3 results use tables that differ from both the printed Eq. (A.6) and the derivation.** The FEMFFUSION tables reproduce the paper's SDP3 eigenvalue; the printed ones are 24 pcm off, and `GARNET`'s re-derived ones 10 pcm (Brantley–Larsen) to 14 pcm (C5G7) lower. `GARNET` keeps the re-derived tables. No code change is needed; the difference is documented in `garnet/spn.py`.

3. **The vacuum default was inconsistent across the family; Marshak is now the default.** When this study started, only the dedicated SP3 solver defaulted to the coupled Marshak condition. SP5, SP7 and the $\mathrm{SDP}_N$ solvers defaulted to the per-moment Robin term, which is 52–73 pcm low on Brantley–Larsen and depends on the moment variables it is applied to (0.35–0.7 pcm between equivalent solvers); on a reflected core such as C5G7 the choice is irrelevant (0.3 pcm). Following this study, every $\mathrm{SP}_N/\mathrm{SDP}_N$ eigen, transient and noise solver applies Marshak by default wherever the coupled term exists (a Cartesian grid with a vacuum face and no hybrid mask) and Robin elsewhere; `marshak_vacuum=False` restores Robin for comparison. This moves SP5, SP7, SDP2 and SDP3 results on problems with vacuum faces, by about +53 pcm on Brantley–Larsen. SDP2 pays for it: it loses its CG path and runs GMRES, 5.8 times slower at $n = 160$ (Table 7); SDP3 already ran GMRES, and $\mathrm{SP}_N$ keeps CG. The dedicated SDP1 solver, eigen and transient, keeps Robin, because the coupled matrix is not symmetric in its two-moment basis; the general $\mathrm{SDP}_N$ block of order 1 gives SDP1 with Marshak. Very large $\mathrm{SDP}_N$ meshes are subject to finding 5.

4. **The C5G7 builder's sampled pin areas were biased; exact areas are now the default.** In the pin-resolved model, each cell that the $r = 0.54$ cm fuel circle cuts is a blend of pin and water weighted by its covered fraction, so a pin's fuel inventory is the sum of those fractions. By default each fraction is estimated by counting which of $16 \times 16$ sample points in the cell fall inside the circle. All 1056 fuel pins are cut identically, so this error does not average out over the core:

it is a systematic error in the fuel inventory, of irregular sign and size, $-0.66\%$ at $s = 4$, $+0.12\%$ at 6, $+0.06\%$ at 8 and $-0.02\%$ at 16. In this lattice $k$ moves by about $-35$ pcm per $+0.1\%$ of fuel area, so the sampled model is 226 pcm high at $s = 4$ and 22 pcm low at $s = 8$, more than any difference between methods in this study and enough to make a mesh study irregular. With `exact_area=True` each fraction is the closed-form area of the square–disk intersection (the routine of the HP-MR fix, audit item O6), and the fuel inventory equals $\pi r^2$ per pin to round-off on every mesh. Following this study, exact areas are the default of the pin-resolved C5G7 and C5G7-TD builders; `exact_area=False` restores the sampling to reproduce older results. No test or stored result depended on the sampled default (the colorset ADF study samples $256 \times 256$ points explicitly); the change moves the output of the examples `c5g7_meshconv.py`, `sdpn_c5g7.py --pin-resolved` and `c5g7_td.py --pin-resolved`. The builder's zero-flux outer faces are within 1.4 pcm of vacuum and stay the default.

5. **The inner solves do not scale to the finest 2-D meshes.** With Marshak vacuum, SDP1, SDP2 and SDP3 at C5G7 $s = 32$ (2.7 million cells) all stopped in the first outer iteration: GMRES with the default Jacobi preconditioner did not reach its tolerance in 5000 iterations (residual 1.3–4.8 times the target). Up to $s = 24$ they converged, but with 121–161 outer iterations against 57–73 for the CG-solved $\mathrm{SP}_N$ blocks. The performance study (Section 5) hit the same cap on Brantley–Larsen, on both devices: GMRES at $n = 1280$ (SDP1, SDP3; 1.6 million cells), and also CG at $n = 2560$ (SP1, SP3; 6.6 million cells). Extruded 3-D meshes are better conditioned at the same cell count and converge up to 105 million cells, but GMRES there is close enough to the cap that two extruded Brantley–Larsen cases converged on the GH200 and failed on the CPU, whose Gram–Schmidt variant rounds differently. Tracked as audit item O14. *Recommendation:* precondition the large non-symmetric blocks (for example `spn_precondition`, already available) and decide whether an inner solve that misses its tolerance in an early outer iteration should abort the eigenvalue solve.

6. **Mesh requirements.** Brantley–Larsen on the $80 \times 80$ test mesh carries $+87$ to $+132$ pcm of mesh error. Pin-resolved C5G7 needs at least 16 cells per pin side for a 4 pcm eigenvalue; at 4 cells it is 68–216 pcm off, and its convergence is not monotone.

7. **C5G7 offset.** `GARNET`'s pin-resolved C5G7 eigenvalues are 11–22 pcm below the paper's, growing with angular order. The models differ in the pin-boundary representation, which the paper does not specify; the offset is 3–5% of the angular error and is left unexplained.

8. **The C5G7 regression test's agreement with MCNP is a cancellation** of $+380$ to $+660$ pcm homogenization error and about $-225$ pcm mesh error. Its docstrings now say so; its envelopes are kept.

9. **Devices agree.** CPU and GH200 give the same $k$ to $1.3 \times 10^{-13}$ in the 18 configurations of the accuracy study, and within $7 \times 10^{-9}$ on every mesh of the performance study, inside the $2 \times 10^{-8}$ allowed between equivalent configurations.

10. **The GH200 pays off from about $10^5$ unknowns.** On identical meshes one GH200 is faster than one CPU core from 0.1–0.2 million unknowns on Brantley–Larsen and from the coarsest C5G7 mesh (except SP3). On the largest 2-D meshes both devices ran it is 22–162 times faster, and the speedup is still rising there. It solves extruded, 3-D-sized versions of both problems with up to 597 million unknowns in 2–34 minutes. On the extruded meshes with 64 layers it is 350–400 times faster than one core for SP1 and SP3 (measured), and about 200–260 times for SDP1 and SDP3 (projected from measured per-iteration costs). Its limits are device memory, reached by $\mathrm{SDP}_N$ at about 450 million unknowns, and the iteration cap of finding 5. Two single-core runs and two controls were still running on 2 November. *Recommendation:* profile SP3's operator apply on the GPU, whose fixed cost per iteration is five to six times SP1's, probably in the coupled Marshak boundary term.

# A Reproduction

From the repository root, with `GARNET` importable (`PYTHONPATH=.`):

```
JOBS=8 bash benchmark-results/carreno2024-sdpn-vv/run_sweep.sh all   # CPU runs
```

```
bash slurm/stage_and_submit_carreno_gh.sh                              # GH200 runs
GARNET_VV_STAGE_ROOT=<new dir> GARNET_VV_ROBIN_CELLS= \
  GARNET_VV_CELLS="4 6 12 20 32 48" GARNET_VV_METHODS="SP1 SP3 SP5 SP7" \
  bash slurm/stage_and_submit_carreno_gh.sh                            # GH200 batch 2
#   (and GARNET_VV_CELLS="4 6 12 20 32" GARNET_VV_METHODS="SDP1 SDP2 SDP3")
bash benchmark-results/carreno2024-sdpn-vv/run_timing.sh             # timings
bash slurm/stage_and_submit_carreno_perf.sh               # GPU/CPU, 2-D
GARNET_PERF_WHICH="gpu-bl3d gpu-c5g7-3d gpu-3d-z32 cpu-bl3d cpu-c5g7-3d" \
  GARNET_PERF_WORK_ROOT=/data/scratch/shared/poncet_m/garnet-carreno-perf3d \
  bash slurm/stage_and_submit_carreno_perf.sh             # GPU/CPU, extruded
bash benchmark-results/carreno2024-sdpn-vv/collect_perf.sh
cd benchmark-results/carreno2024-sdpn-vv && make
```

`tools/assess_carreno2024.py` writes one JSON record per run: $k$, convergence, outer and inner iterations, setup and solve seconds, the flux on the paper's plotted lines and, for C5G7, the $51 \times 51$ pin powers. `data/runs/` holds the CPU records (merlin7 `cn000`, one thread per run, up to 16 runs at a time; `progress.log` lists them, including the runs stopped as too slow), `data/runs-gh200/` the GH200 records with their job logs and staging provenance (two batches, both commit `20967d3` plus the uncommitted audit work), and `data/timing/` the best-of-3 timings, taken with nothing else of this study running. The SDP3 table variants (`SDP3-A6`, `SDP3-FFN`) were run with the same tool outside the sweep. `data/paper-figures.json` is the output of `extract_paper_figures.py` (it needs PyMuPDF; the report build does not). `data/nea-c5g7-mcnp-pin-powers.json` is the NEA MCNP pin-power map as distributed with FEMFFUSION (`examples/2D_C5G7/c5g7_transport.ref`, SHA-256 recorded in the file); its assembly powers are 492.80, 211.71 and 139.78 and its largest pin power 2.498, the NEA values. `make_figures.py` turns all of these into `figures/`, `tables/` and `data/summary.json`.